



Society of Petroleum Engineers

SPE-226860-MS

Hybrid Nanoparticles for Enhanced Assembly at Oil-Water Interfaces: A CO₂-Assisted Functionalization Approach

Ahmed Wasel Alsmail, Expec Advanced Research Center, Saudi Aramco, Saudi Arabia; Emmanuel P. Giannelis, Material Science and Engineering Department, Cornell University, United States

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226860-MS](https://doi.org/10.2118/226860-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This research presents a novel approach to enhance the assembly of nanoparticles (NPs) at the oil-water interfaces through hybrid functionalization. Silica NPs were initially functionalized with two silanes, which improved their colloidal stability at high temperature and salinity environment. However, at neutral pH, the NPs failed to assemble at the interface due to particle-particle repulsive interactions. To address this limitation, a third functionalization step was introduced, involving the incorporation of stearic acid into the NP surface via an interfacial assembly method facilitated by CO₂ bubbling, which enabled efficient dispersion of the particles and provided additional binding sites for the stearic acid molecules.

The resulting hybrid NPs exhibited enhanced hydrophobicity, leading to improved assembly at the oil-water interface. Interfacial tension measurements revealed a significant reduction in interfacial tension, comparable to values obtained with molecular surfactants. Additionally, the hybrid NPs displayed increased affinity towards the interface, driven by hydrophobic interactions. Contact angle measurements confirmed the improved hydrophobicity of the hybrid NPs, showing a 30-degree reduction in contact angle compared to the original NPs.

Transmission Electron Microscopy (TEM) and X-Ray Photoelectron Spectroscopy (XPS) analyses validated the successful incorporation of stearic acid onto the NP surface. Thermogravimetric Analysis (TGA) showed a notable increase in organic content, indicating the attachment of stearic acid molecules. These findings demonstrate the effectiveness of the proposed hybrid functionalization strategy in overcoming the limitations of traditional NP-based systems.

The development of these hybrid NPs opens up opportunities for exploiting them in complex environments, such as those encountered in industrial processes, biomedical applications, and environmental remediation. By leveraging the unique properties of these hybrid NPs, researchers can design and engineer novel systems capable of manipulating fluid-fluid interfaces, enabling breakthroughs in fields like emulsion stabilization, microfluidics, and nanotechnology.

Introduction

Functionalizing nanoparticles, silica nanoparticles in particular, represents a crucial step in tailoring their behavior by adjusting both inter- and intramolecular interactions. A commonly employed method involves the use of various silanes, which react with particles through surface hydroxyl groups. This interaction imparts distinct surface chemistry and properties to the particles. While there is considerable information on surface modification using single silanes, the literature appears relatively sparse when it comes to the concurrent use of more than one silane for functionalization.(Fernandes et al., 2014; Kang et al., 2019; Karnati et al., 2020; Knowles et al., 2020; Lecot et al., 2020; Shimada et al., 2003; Wang et al., 2021)

In addition, the assembly of NPs at oil-water interfaces (referred to as Pickering emulsions) and their ability to manipulate liquid-liquid interfaces has been reviewed extensively.(Aldakkan et al., 2021; Alsmail et al., 2021; Goldhahn et al., 2018; Guzmán et al., 2022; Hammami et al., 2021; Lee et al., 2018; Mohammed et al., 2020, 2022; Vasantha et al., 2020; Yang et al., 2011; F. Zhang et al., 2019),(Guzmán et al., 2022; Shi & Russell, 2018)

It was demonstrated that the double silane-functionalized silica nanoparticles exhibit enhanced colloidal stability, especially in the presence of various electrolytes. Utilizing these colloidally stable silica nanoparticles, the assembly mechanism at oil-water interfaces through static- and time-dependent measurements both as a function of pH and electrolytes, incorporating in operando ultrasmall-/small-angle X-ray scattering (USAXS/SAXS) was investigated. The study provided fundamental insights into the assembly's mechanistic details, which, in turn, are correlated with the interfacial mechanical properties in tension as well as in shear.(Alsmail, Kouloumpis, et al., 2023; Alsmail, Mohammed, et al., 2023)

One of the limitations of this system is that at pH 7, due to particle-particle repulsive interactions, the assembly is not established and the stability of emulsion deteriorates. Moreover, the interfacial tension with nonpolar oil is almost unchanged in the presence of the particles. One of the reasons for this observation is that particles pinned at the oil-water interface experience a long-range repulsive dipole-dipole interaction due to the asymmetric counter-ion distribution between oil and water. Specifically, inside the double layer of the NPs, excess counterions screen the electric field from the particle surface. However, there are no mobile ions in the oil phase and the field could be further extended. The asymmetry results in a normal dipole to the interface. As the particles come closer to each other, the double layer would overlap and the interactions result in a strong short-range repulsion that decays exponentially with separation distance. Particles trapped at the oil-water interface develop an electric double layer in the water side, which decays to zero away from the surface of the particle. However, the electric double layer extends a lot more on the oil side.(Uppapalli & Zhao, 2014) This phenomenon explains the reason why the charged double functionalized nanoparticles ($S_{1,2}$) do not assemble at the nonpolar oil-water interface in the absence of electrolytes.

To overcome this limitation, without the need of using electrolytes to screen the repulsive interactions between the particles, and to expand the functionalization possibilities of these nanoparticles, a secondary functionalization step was introduced. This approach accomplished through an interfacial assembly method, improves the hydrophobicity of the particles and leads to hybrid nanoparticles.

In the past, synthesis of Janus particles have been achieved using three main methods: phase separation, masking and self-assembly. One of the most utilized methods is the use of Pickering emulsions where one half of the spherical particle is arrested in the water phase and the other is in the oil phase. One of the phases contains the additional molecules, either hydrophilic (in the water phase) or hydrophobic (in the oil phase). The oil phase utilized is normally wax. In this method, after forming the wax-in-water emulsions, the mixture is cooled down so the wax freezes. After the functionalization, the mixture is heated to melt the wax and extensive cleaning of the excess wax and unreacted materials is necessary.(Kaewsaneha et al., 2013; Perro et al., 2009; Su et al., 2019; J. Zhang et al., 2017) A summary of this widely utilized method is summarized in Figure 1.

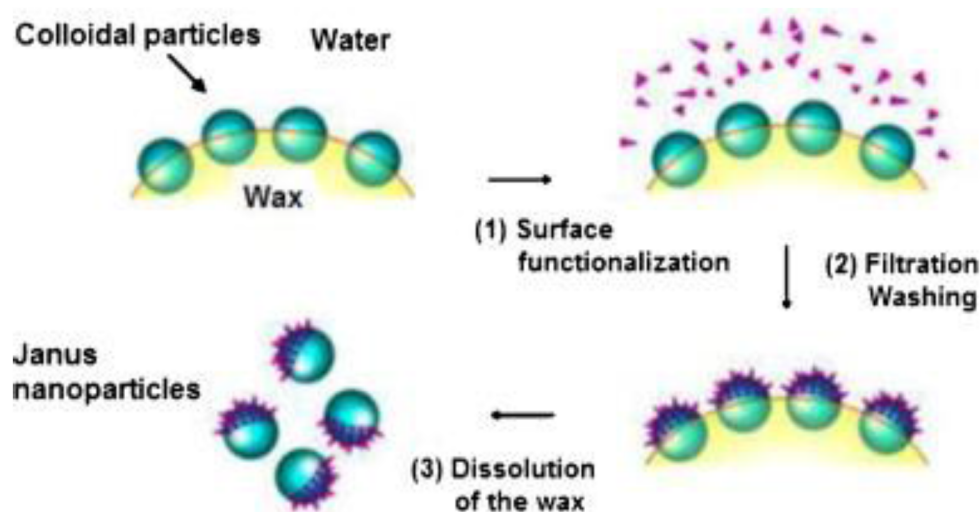


Figure 1—Schematic representation of the preparation of Janus particles utilizing wax-in-water emulsions.(Perro et al., 2009)

The main advantage of utilizing wax as the oil phase is that upon cooling down, the particles are locked in their position.

In contrast, in the approach described below interfacial jamming of particles is used. Particles jammed at the interface lose their mobility and hence make their functionalization through one of the phases a possible and a more reliable approach.(Cheng & Velankar, 2009) This innovative approach not only broadens the scope of applications for these particles but also introduces a new dimension to their interfacial behavior. The incorporation of stearic acid, with its distinctive hydrophobic properties, is anticipated to impart hydrophobic characteristics to the nanoparticles, introducing a novel approach to their assembly at oil-water interfaces. This next phase aims to advancing the understanding of complex interfacial phenomena and engineering nanoparticles with tailored properties for a spectrum of applications.

Experimental Section

Synthesis Procedure

A certain amount of stearic acid (1 mg/mL or 5 mg/mL) was dissolved in hexane by stirring at 50 °C for 1 hour. The $S_{1,2}$ particles were diluted to 0.5 wt%, brought to pH 9 by adding 1 M NaOH and sonicated for 1 hour in a water bath sonicator. Ice was added periodically to prevent heating of the sample. Equal volume of the particles and the hexane containing stearic acid was homogenized at 10,000 rpm for 30 minutes using an IKA Ultra-Turrax T-25 with an 8 mm dispersion tool. The emulsion was then bubbled with CO_2 gas for 5 minutes to protonate the amines, which facilitates dispersion of the particles, as well as provide more binding sites for the stearic acid molecules. The hexane was then allowed to evaporate at room temperature for 48 hours. To remove the excess unreacted stearic acid molecules, the mixture was washed repeatedly with ethanol. The final product was then dried and redispersed in DI water for further characterization. A schematic representation of the synthesis procedure is shown in Figure 2. $S_{1,2,1}$ and $S_{1,2,5}$ refer to the particles where 1 and 5 mg/mL of stearic acid was used in the oil phase.

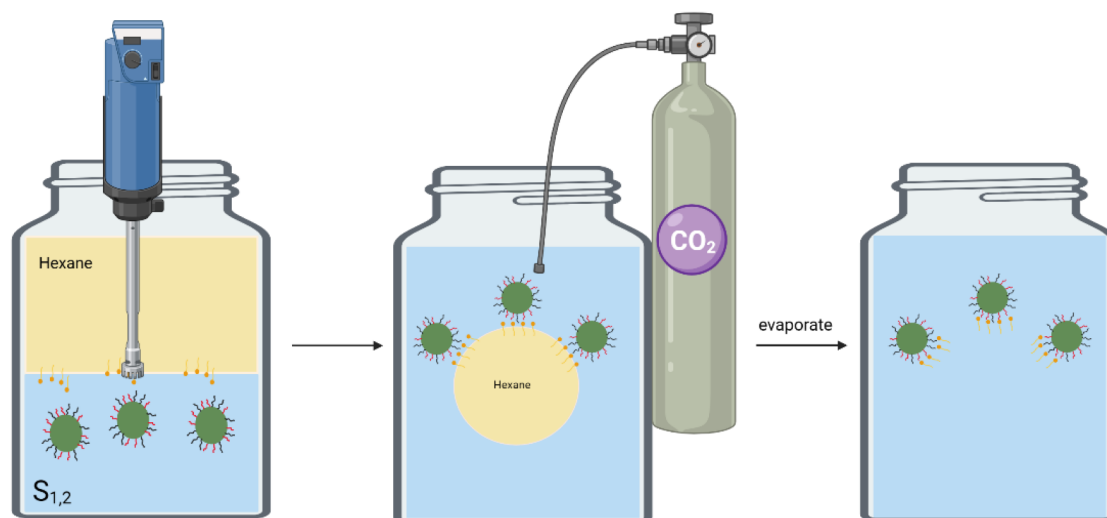


Figure 2—Schematic representation of the synthesis of the hybrid particles using interfacial assembly approach. The orange symbols indicate stearic acid, the red and black lines are the S_1 and S_2 molecules and the green particles are the colloidal silica particles.

Contact angle measurements were performed using the sessile drop technique (Biolin Scientific, Theta Lite). A glass slide was made hydrophobic by immersion in an oil bath at 80 °C for 4 days. The glass slide was then washed with water and isopropanol to remove the excess oil and dried in an oven at 90 °C. A droplet of either deionized water, $S_{1,2}$ or $S_{1,2,5}$ suspension was then placed over the glass slide to determine the contact angle.

Zeta-potential was determined at 25 °C using the Zetasizer ZS90 (Malvern Instruments Ltd., U.K.) and Malvern polycarbonate folded capillary cells (DTS1070) with backscattering detection at 173°. The samples were allowed to equilibrate in the instrument chamber for 120 s prior to measurement. The suspension (1 wt.%) was sonicated for 30 min before each measurement using a water bath sonicator.

The morphology and size of the particles were studied using Transmission Electron Microscopy (FEI-Tecna 12 BioTwin TEM, FEI company; Hillsboro OR) equipped with an LaB6 electron source at 120 kV using a Gatan Orius S1000 CCD camera. A 0.1 wt.% suspension in water was placed onto a carbon coated copper grid prior to imaging. The particle size distribution was analyzed using ImageJ software.

X-ray photoelectron spectroscopy (XPS) was performed using an ESCA 2SR XPS equipped with an Al Ka 1486.7 eV monochromator and an operating pressure of 10^{-9} Torr. The samples in powder form were deposited on copper tape. Survey and high-resolution spectra were collected at 200 eV pass energy and 50 eV pass energy, respectively. Acquired spectra were corrected using the C1s core level region as reference.

Interfacial tension and dilatational modulus were measured using a Spinning Drop Tensiometer (SDT, KRÜSS). An oil droplet was injected in a capillary tube with an inner diameter of 3.25 mm and rotated at fixed speed. The rotational speed was set at 10,000 rpm for all experiments. The oil droplet shape was fitted to the Young-Laplace equation to determine the IFT. (Viades-Trejo & Gracia-Fadrique, 2007) In the early stages of the experiment, the IFT decreases continuously as the droplet rotated and it reaches a plateau after some time (the plateau value is defined as the equilibrium IFT).

Results And Discussion

Figure 3 shows the interfacial tension of 0.5 wt% $S_{1,2}$ suspension with 1 mg/mL and 5 mg/mL stearic acid in hexane. The IFT of the hexane-water system ranges between 49-52 mN/m. (Saïen et al., 2014) The addition of stearic acid and NPs at pH 9 resulted in significant decrease of the IFT to 7 mN/m after 1,100 s of spinning the oil droplet against the suspension at 10,000 rpm. With increasing concentration of stearic acid, the IFT further reduced to an ultralow value of 0.09 mN/m, similar to that reported for molecular surfactants.

(Noorizadeh Bajgirani & Saeedi Dehaghani, 2023; Yekeen et al., 2020) The initial IFT was already less than 1 mN/m due to the fast adsorption of the particles at the interface, which could not be captured with the spinning drop tensiometer. The IFT of the 5 mg/mL SA in hexane was 37 mN/m whereas that of the 1 mg/mL was 42 mN/m suggesting that the reduction of the IFT is mainly due to the presence of the NPs at the interface. When the oil droplet was brought to rest, due to the strong electrostatic interaction of the positively charged particles and the negative oil-water interface, the oil droplet jammed as shown in the onset of Figure 3. The smaller volume utilized for the 5 mg/mL SA is due to the elongation of the oil droplet. This resulted in the droplet edges to be out of the view of the measurement.

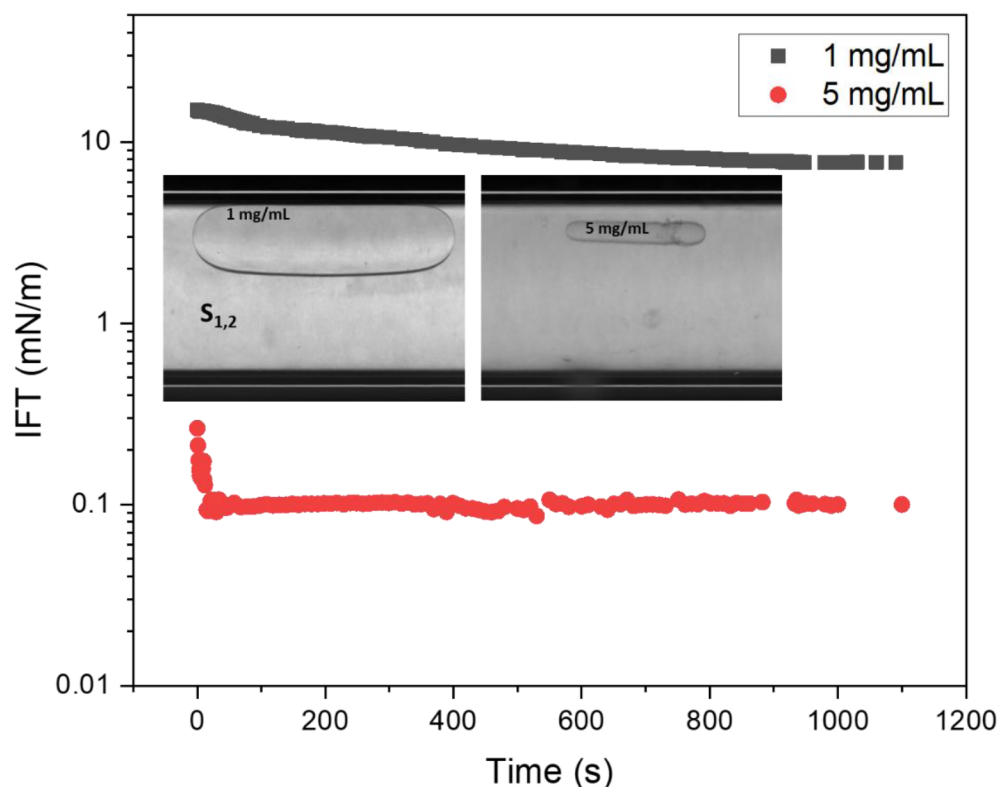


Figure 3—The interfacial tension of 1 mg/mL (black) and 5 mg/mL (red) stearic acid in hexane and 0.5 wt% suspension at pH 9. The onset is the jammed oil droplet at the end of the experiment.

The dilatational modulus of the system at 1 mg/mL was 650 ± 50 mN/m suggesting that a solid-like interface was formed. Similarly, when the stearic acid was increased to 5 mg/mL, the dilatational modulus increased to 720 ± 30 mN/m.

The high-resolution XPS of C1s further confirms the successful incorporation of the stearic acid on the surface of the particles (Figure 4). The $S_{1,2}$ particles only contain C-O (from atmospheric CO_2) and C peaks that come from the silanes (S_1 and S_2). However, for the hybrid particles, a new peak emerges at 289 eV. This peak is attributed to the carboxylic acid group. (Artemenko et al., 2021) Also, the carbon content increases by 8 % from that of the $S_{1,2}$.

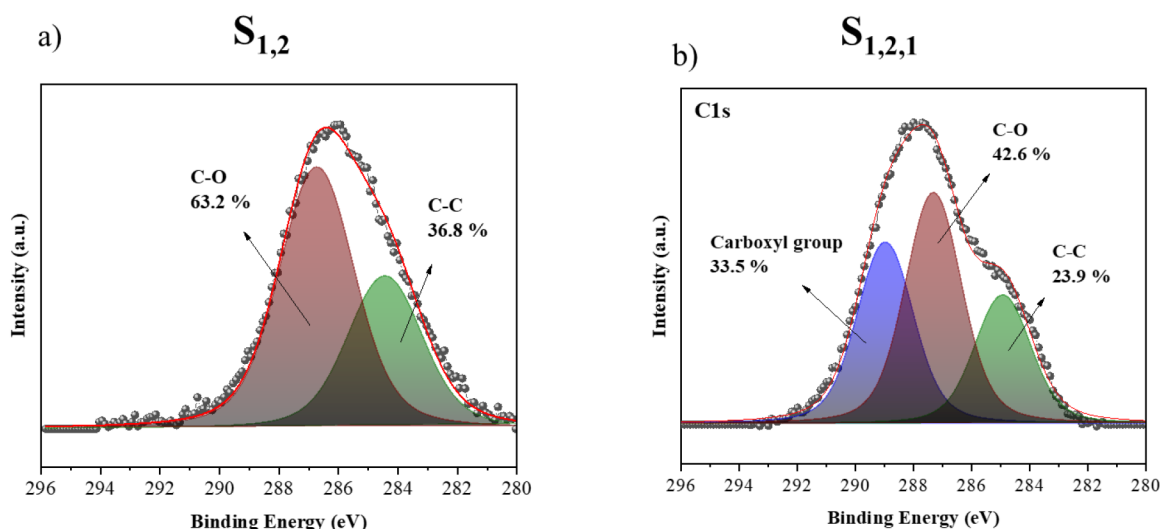


Figure 4—C1s XPS high resolution spectra for the $S_{1,2}$ (a) and the hybrid $S_{1,2,1}$ particles (b)

The TEM images shown in Figure 5a indicate that after evaporating the hexane, there is a closed-packed assembly of particles around open areas which were most likely the oil phase. After cleaning the particles and resuspending, the average size detected with TEM was 16.5 nm in comparison to 15 nm for the $S_{1,2}$ particles. The smaller increase in the size further supports the incorporation of the stearic acid into the $S_{1,2}$ particles.

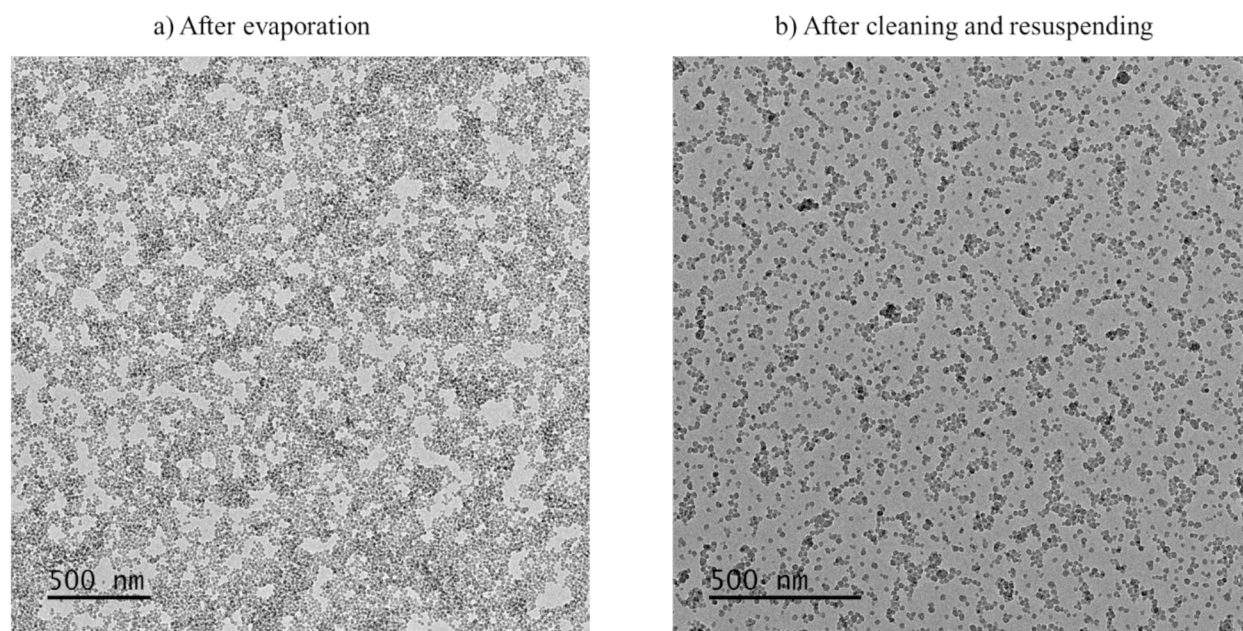


Figure 5—TEM images of a) after evaporation of the hexane from the emulsion and b) the $S_{1,2,1}$ particle after cleaning and resuspending

The TGA analysis shown in Figure 6 compares the weight loss for the $S_{1,2}$, $S_{1,2,1}$ and $S_{1,2,5}$. It should be mentioned that the thermal decomposition of stearic acid takes place between 200-300 °C.(Zhu et al., 2021) The pristine particles have no weight loss from 200-300 °C. For the hybrid particles when 1 mg/mL was used in the oil phase, the weight loss in that region is about 1%. The corresponding loss for the system prepared with 5 mg/mL is 30% and is attributed to the loss of stearic acid. For the $S_{1,2,1}$ sample, the molar ratio of the stearic acid to silanes is 0.1, which is reasonable given the low starting amount of stearic acid. In

contrast, when the oil phases contains 5 mg/mL, the ratio increases to 1.25 moles/mole. With that in mind, one could state that the silanes used in the reaction have 1 primary amine and 1 quaternary ammonium that are exposed on the surface. Hence, for each 1 mole of silane, we have two moles of nitrogen and the final ratio of stearic acid/Nitrogen in moles/moles is 1.25/2. If the total number of nitrogen is considered the ratio is then 1.25/4.

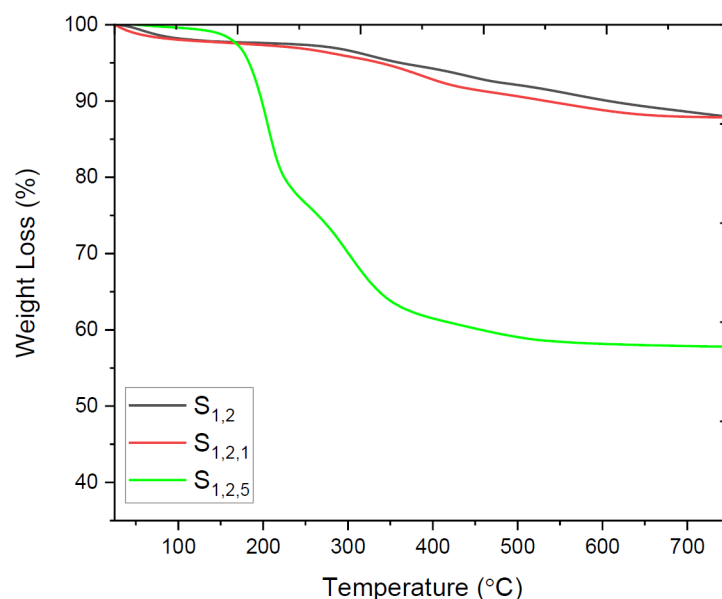


Figure 6—TGA analysis of the S_{1,2} (black), hybrid S_{1,2} with 1mg/mL starting stearic acid (red) and hybrid S_{1,2} with 5mg/mL starting stearic acid (green)

As mentioned previously, one of the main advantages of this approach is to drive the particles adsorption to a nonpolar oil-water interface. To test this assumption, dodecane was used as the nonpolar oil phase. As shown in Table 1, in the presence of the S_{1,2} particles, the IFT reduces only slightly in comparison to that of the pristine dodecane-water interface (46 mN/m in comparison to 50 mN/m). However, for the S_{1,2,1} which contains a minute amount of SA, the IFT reduces significantly to 22 mN/m. When the more hydrophobic S_{1,2,5} particles were used the IFT reduced even further to 12 mN/m. The preferential adsorption of the modified particles to the interface is explained by the increase in the hydrophobic interactions. (Bizmark et al., 2014) This was already seen with the reduction of the effective zeta-potential of the particles with increasing amount of stearic acid (Table 1). The screening of the particle-particle repulsion by incorporating stearic acid into the surface of the S_{1,2} enabled more extensive packing of the particles at the interface and lower interfacial tension due to the interaction between the hydrophobic tail of the stearic acid with dodecane, similar to molecular surfactant. However, for the pristine S_{1,2} particles, the assembly is limited due to the particle-particle repulsions and image-charge repulsion as shown schematically in Figure 7. (Leunissen et al., 2007)

Table 1—Summary of the results of the S_{1,2} and the hybrid S_{1,2} particles.

System	Zeta-potential (mV)	Organic Content (%)	SA/S _{1,2} moles/mole	IFT with Dodecane (mN/m)
S _{1,2}	32±2	10	0	46±1
S _{1,2,1}	20±3	12	0.1	22±3
S _{1,2,5}	15±2	40.8	1.25	12±2

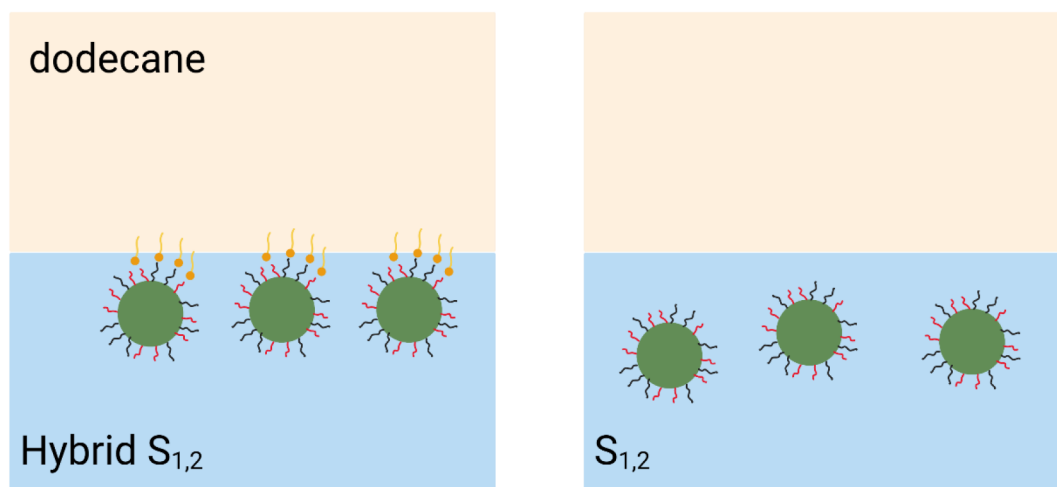


Figure 7—Schematic representation of the assembly of the particles at the dodecane-oil water interface. In case of the hybrid particles, the assembly is possible through the hydrophobic interactions. However, for the pristine $S_{1,2}$ particles, the assembly is limited due to the particle-particle repulsions and image-charge repulsion.

The contact angle measurements further confirm the improved hydrophobicity of the nanoparticles after incorporating stearic acid. As shown in Figure 8, when a droplet of water is added, the contact angle is 93° . A slight reduction of the contact angle takes place when a suspension of the $S_{1,2}$ particle is placed. The $S_{1,2}$ particles only reduced the contact angle slightly from 93° to 89° . However, improved spreading of the droplet is seen when a droplet containing the $S_{1,2,5}$ particles, which are partially hydrophobic, is placed on the substrate. The contact angle is reduced to 61° .

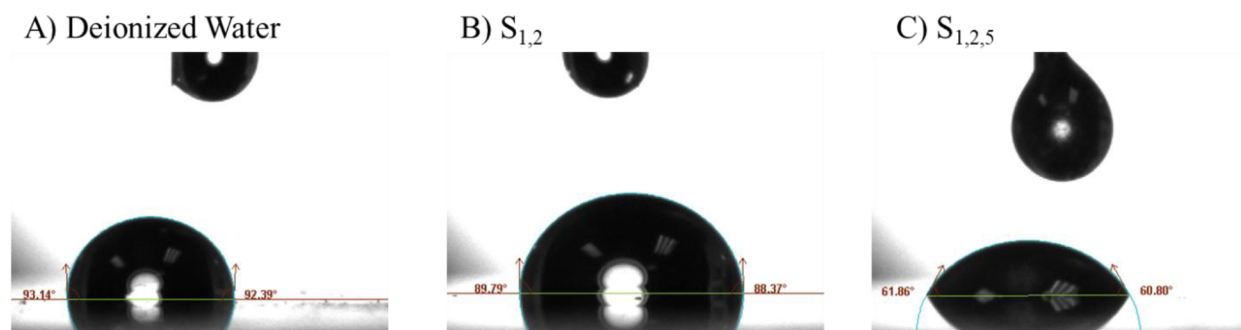


Figure 8—The contact angle measurement of A) a water droplet, B) $S_{1,2}$ suspension and C) $S_{1,2,5}$ suspension

Conclusion

In summary, while dual silane functionalization of silica NPs enhanced their colloidal stability, challenges at pH 7 vis a vis the limited impact on interfacial tension necessitated a different approach using *hybrid functionalization* of NPs. The new approach involves first the functionalization of the silica with the two silanes as in 1:1 ratio as described in the previously followed by a selective decoration of the particles by stearic acid after assembly at an oil-water interface with the oil phase containing different amounts of stearic acid. The process results in more hydrophobic NPs. The resulting NPs exhibited reduced interfacial tension and enhanced assembly potential at the nonpolar oil-water interface. Moreover, the contact angle of a hydrophobic surface was lower by 30° when utilizing the hydrophobized particles in comparison to the $S_{1,2}$ particles.

The preferential adsorption of the hybrid NPs at the interface, driven by hydrophobic interactions, provides a solution to the limitations faced by the previous NPs and offers the means to use NPs in complex environments for a number of practical applications.

References

- Aldakkan, B. S., Hammami, M. A., Qi, G., Kanj, M. Y., & Giannelis, E. P. (2021). Stimuli-Responsive, Hydrolyzable Poly(Vinyl Laurate-co-vinyl Acetate) Nanoparticle Platform for In Situ Release of Surfactants. *ACS Applied Materials & Interfaces*, **13**(21), 25553–25562. <https://doi.org/10.1021/acsami.1c04977>
- Alsmacil, Ahmed Wasel. *Colloidal Stability and Assembly of Double Functionalized Silica Nanoparticles at Oil-Water Interfaces: An Approach to Interfacial Engineering Strategies*. Diss. Cornell University, 2024.
- Alsmacil, A. W., Hammami, M. A., Enotiadis, A., Kanj, M. Y., & Giannelis, E. P. (2021). Encapsulation of an Anionic Surfactant into Hollow Spherical Nanosized Capsules: Size Control, Slow Release, and Potential Use for Enhanced Oil Recovery Applications and Environmental Remediation. *ACS Omega*, **6**(8), 5689–5697. <https://doi.org/10.1021/acsomega.0c06094>
- Alsmacil, A. W., Kouloumpis, A., Potsi, G., Hammami, M. A., Kanj, M. Y., & Giannelis, E. P. (2023). Probing the Interfacial Properties of Oil–Water Interfaces Decorated with Ionizable, pH Responsive Silica Nanoparticles. *Langmuir*, **39**(8), 3118–3130. <https://doi.org/10.1021/acs.langmuir.2c03286>
- Alsmacil, A. W., Mohammed, S., Aldakkan, B. S., Chalmpes, N., Kouloumpis, A., Potsi, G., Galvin, A., Gadikota, G., Kanj, M. Y., & Giannelis, E. P. (2023). Fine-Tuning the Surface and Interfacial Chemistry of Silica Nanoparticles to Control Stability in High Ionic Strength Electrolytes and Modulate Assembly at Oil–Water Interfaces. *Chemistry of Materials*, **35**(21), 9150–9159. <https://doi.org/10.1021/acs.chemmater.3c01871>
- Artemenko, A., Shchukarev, A., Štenclová, P., Wågberg, T., Segervald, J., Jia, X., & Kromka, A. (2021). Reference XPS spectra of amino acids. *IOP Conference Series: Materials Science and Engineering*, **1050**(1), 012001. <https://doi.org/10.1088/1757-899X/1050/1/012001>
- Bizmark, N., Ioannidis, M. A., & Henneke, D. E. (2014). Irreversible Adsorption-Driven Assembly of Nanoparticles at Fluid Interfaces Revealed by a Dynamic Surface Tension Probe. *Langmuir*, **30**(3), 710–717. <https://doi.org/10.1021/la404357j>
- Cheng, H.-L., & Velankar, S. S. (2009). Controlled Jamming of Particle-Laden Interfaces Using a Spinning Drop Tensiometer. *Langmuir*, **25**(8), 4412–4420. <https://doi.org/10.1021/la803552j>
- Fernandes, N. J., Wallin, T. J., Vaia, R. A., Koerner, H., & Giannelis, E. P. (2014). Nanoscale Ionic Materials. *Chemistry of Materials*, **26**(1), 84–96. <https://doi.org/10.1021/cm402372q>
- Goldhahn, C., Schubert, J., Schlaad, H., Ferri, J. K., Fery, A., & Chanana, M. (2018). Synthesis of Metal@Protein@Polymer Nanoparticles with Distinct Interfacial and Phase Transfer Behavior. *Chemistry of Materials*, **30**(19), 6717–6727. <https://doi.org/10.1021/acs.chemmater.8b02314>
- Guzmán, E., Martínez-Pedrero, F., Calero, C., Maestro, A., Ortega, F., & Rubio, R. G. (2022). A broad perspective to particle-laden fluid interfaces systems: From chemically homogeneous particles to active colloids. *Advances in Colloid and Interface Science*, **302**, 102620. <https://doi.org/10.1016/j.cis.2022.102620>
- Hammami, M. A., Kanj, M. Y., & Giannelis, E. P. (2021). Monitoring the Early Stages of Formation of Oil–Water Emulsions Using Flow Cytometry. *Langmuir*. <https://doi.org/10.1021/acs.langmuir.1c01847>
- Kaewsaneha, C., Tangboriboonrat, P., Polpanich, D., Eissa, M., & Elaissari, A. (2013). Preparation of Janus colloidal particles via Pickering emulsion: An overview. *Nanoparticles@interfaces*, **439**, 35–42. <https://doi.org/10.1016/j.colsurfa.2013.01.004>
- Kang, H., Long, D. J., & Haynes, C. L. (2019). Preparation of Colloidally Stable Positively Charged Hollow Silica Nanoparticles: Effect of Minimizing Hydrolysis on ζ Potentials. *Langmuir*, **35**(24), 7985–7994. <https://doi.org/10.1021/acs.langmuir.9b01042>
- Karnati, S. R., Oldham, D., Fini, E. H., & Zhang, L. (2020). Application of surface-modified silica nanoparticles with dual silane coupling agents in bitumen for performance enhancement. *Construction and Building Materials*, **244**, 118324. <https://doi.org/10.1016/j.conbuildmat.2020.118324>
- Knowles, B. R., Wagner, P., MacLaughlin, S., Higgins, M. J., & Molino, P. J. (2020). Modified silica nanoparticle coatings: Dual antifouling effects of self-assembled quaternary ammonium and zwitterionic silanes. *Biointerphases*, **15**(2), 021009. <https://doi.org/10.1116/1.5143141>
- Lecot, S., Lavigne, A., Yang, Z., Géhin, T., Botella, C., Jousseume, V., Chevolot, Y., Phaner-Goutorbe, M., & Yeromonahos, C. (2020). Arrangement of Monofunctional Silane Molecules on Silica Surfaces: Influence of Alkyl Chain Length, Head-Group Charge, and Surface Coverage, from Molecular Dynamics Simulations, X-ray Photoelectron Spectroscopy, and Fourier Transform Infrared Spectroscopy. *The Journal of Physical Chemistry C*, **124**(37), 20125–20134. <https://doi.org/10.1021/acs.jpcc.0c05349>
- Lee, J. G., Larive, L. L., Valsaraj, K. T., & Bharti, B. (2018). Binding of Lignin Nanoparticles at Oil–Water Interfaces: An Ecofriendly Alternative to Oil Spill Recovery. *ACS Applied Materials & Interfaces*, **10**(49), 43282–43289. <https://doi.org/10.1021/acsami.8b17748>

- Leunissen, M. E., van Blaaderen, A., Hollingsworth, A. D., Sullivan, M. T., & Chaikin, P. M. (2007). Electrostatics at the oil–water interface, stability, and order in emulsions and colloids. *Proceedings of the National Academy of Sciences*, **104**(8), 2585–2590. <https://doi.org/10.1073/pnas.0610589104>
- Mohammed, S., Asgar, H., Kuzmenko, I., & Gadikota, G. (2020). Self-Assembly of Silica Nanoparticles at Water–Hydrocarbon Interfaces: Insights from In Operando Small-Angle X-ray Scattering Measurements and Molecular Dynamics Simulations. *Energy & Fuels*, **34**(10), 12545–12555. <https://doi.org/10.1021/acs.energyfuels.0c02759>
- Mohammed, S., Kuzmenko, I., & Gadikota, G. (2022). Reversible assembly of silica nanoparticles at water–hydrocarbon interfaces controlled by SDS surfactant. *Nanoscale*, **14**(1), 127–139. <https://doi.org/10.1039/D1NR06807E>
- Noorizadeh Bajgirani, S. S., & Saeedi Dehaghani, A. H. (2023). Experimental investigation of wettability alteration, IFT reduction, and injection schemes during surfactant/smart water flooding for EOR application. *Scientific Reports*, **13**(1), 11362. <https://doi.org/10.1038/s41598-023-37657-1>
- Perro, A., Meunier, F., Schmitt, V., & Ravaine, S. (2009). Production of large quantities of "Janus" nanoparticles using wax-in-water emulsions. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, **332**(1), 57–62. <https://doi.org/10.1016/j.colsurfa.2008.08.027>
- Saien, J., Rezvani Pour, A., & Asadabadi, S. (2014). Interfacial Tension of the n-Hexane–Water System under the Influence of Magnetite Nanoparticles and Sodium Dodecyl Sulfate Assembly at Different Temperatures. *Journal of Chemical & Engineering Data*, **59**(6), 1835–1842. <https://doi.org/10.1021/je401066j>
- Shi, S., & Russell, T. P. (2018). Nanoparticle Assembly at Liquid–Liquid Interfaces: From the Nanoscale to Mesoscale. *Advanced Materials*, **30**(44), 1800714. <https://doi.org/10.1002/adma.201800714>
- Shimada, T., Aoki, K., Shinoda, Y., Nakamura, T., Tokunaga, N., Inagaki, S., & Hayashi, T. (2003). Functionalization on Silica Gel with Allylsilanes. A New Method of Covalent Attachment of Organic Functional Groups on Silica Gel. *Journal of the American Chemical Society*, **125**(16), 4688–4689. <https://doi.org/10.1021/ja0346911>
- Su, H., Hurd Price, C.-A., Jing, L., Tian, Q., Liu, J., & Qian, K. (2019). Janus particles: Design, preparation, and biomedical applications. *Materials Today Bio*, **4**, 100033. <https://doi.org/10.1016/j.mtbio.2019.100033>
- Uppapalli, S., & Zhao, H. (2014). The influence of particle size and residual charge on electrostatic interactions between charged colloidal particles at an oil–water interface. *Soft Matter*, **10**(25), 4555–4560. <https://doi.org/10.1039/C4SM00527A>
- Vasanth, V. A., Hua, N. Q., Rusli, W., Hadia, N. J., & Stubbs, L. P. (2020). Unique Oil-in-Brine Pickering Emulsion Using Responsive Antipolyelectrolyte Functionalized Latex: A Versatile Emulsion Stabilizer. *ACS Applied Materials & Interfaces*, **12**(20), 23443–23452. <https://doi.org/10.1021/acsami.0c03743>
- Viades-Trejo, J., & Gracia-Fadrique, J. (2007). Spinning drop method: From Young–Laplace to Vonnegut. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, **302**(1), 549–552. <https://doi.org/10.1016/j.colsurfa.2007.03.033>
- Wang, L., Schubert, U. S., & Hoeppener, S. (2021). Surface chemical reactions on self-assembled silane based monolayers. *Chemical Society Reviews*, **50**(11), 6507–6540. <https://doi.org/10.1039/D0CS01220C>
- Yang, X.-C., Samanta, B., Agasti, S. S., Jeong, Y., Zhu, Z.-J., Rana, S., Miranda, O. R., & Rotello, V. M. (2011). Drug Delivery Using Nanoparticle-Stabilized Nanocapsules. *Angewandte Chemie International Edition*, **50**(2), 477–481. <https://doi.org/10.1002/anie.201005662>
- Yekeen, N., Padmanabhan, E., Syed, A. H., Sevo, T., & Kanesen, K. (2020). Synergistic influence of nanoparticles and surfactants on interfacial tension reduction, wettability alteration and stabilization of oil-in-water emulsion. *Journal of Petroleum Science and Engineering*, **186**, 106779. <https://doi.org/10.1016/j.petrol.2019.106779>
- Zhang, F., Yang, F., Hua, M., Yang, Z., Wei, H., Yang, Y., & Wei, J. (2019). Buckling of Two-Dimensional Colloidal Nanoplatelets in Confined Space To Design Heterogeneous Catalysts. *Chemistry of Materials*, **31**(10), 3812–3817. <https://doi.org/10.1021/acs.chemmater.9b01260>
- Zhang, J., Grzybowski, B. A., & Granick, S. (2017). Janus Particle Synthesis, Assembly, and Application. *Langmuir*, **33**(28), 6964–6977. <https://doi.org/10.1021/acs.langmuir.7b01123>
- Zhu, C., Chen, Y., Cong, R., Ran, F., & Fang, G. (2021). Improved thermal properties of stearic acid/high density polyethylene/carbon fiber composite heat storage materials. *Solar Energy Materials and Solar Cells*, **219**, 110782. <https://doi.org/10.1016/j.solmat.2020.110782>